

PCAP Investigation Report - Network Traffic Analysis

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Case ID: PCAP-101

Severity: High

1. Objective

The objective of this investigation was to analyze captured network traffic contained within a PCAP file to identify indicators of compromise (IOCs), anomalous communication patterns, evidence of malware payload delivery, post infection activities or command and control behaviors.

2. Tools and Data Sources

- **Wireshark:** Network traffic analyzer
- **PCAP Source:** Malware-Traffic-Analysis.net (Public Dataset)
- **Threat Intelligence:** urlscan.io , [virustotal.com](https://www.virustotal.com)

3. PCAP Overview

The analyzed PCAP contains network traffic associated with a suspected malware infection scenario. Observed traffic included:

- DNS requests and responses
- TCP sessions between internal hosts and external IP addresses
- HTTP and HTTPS request and response traffic

The analysis focuses on identifying suspicious communication patterns, malware activities, data exfiltration, post-compromise activities or command and control infrastructure.

4. Investigation Methodology

4.1 Protocol Identification

The Investigation started with protocol level filtering to establish a baseline of observed traffic. Primary protocols analyzed included:

- DNS
- TCP
- HTTP/HTTPS

4.2 DNS Analysis

DNS traffic was filtered to identify suspicious domains in the traffic. Multiple suspicious domains were identified that did not align with known legitimate services. These domains were enriched using the open source threat intelligence platforms and were flagged as malicious

4.3 IP Address Analysis

External IP addresses communicating with the internal host after DNS resolution was reviewed. Indicators observed included:

- Repeated structured outbounds connections to same external IP addresses
- Multiple TCP communications with the external IP addresses were observed which is consistent with command and control behavior.

Threat intelligence enrichment confirmed that several of these IP addresses are associated with malicious infrastructure.

4.4 HTTP and HTTPS Review

HTTP and HTTPS traffic was filtered and analyzed. The observed indicators are:

- Multiple POST and GET requests were made to the identified malicious domains, which indicate potential malware payload delivery or command and control instructions.
- GET request was made to a malicious domain, deep packet analysis confirmed the request was made from the victim Powershell terminal and further analysis proves the request contains obfuscated powershell codes which could be a potential payload delivery or instructions from the malicious server.
- Multiple structured POST requests with 30s intervals to multiple malicious domains were observed after the GET requests. The pattern is consistent with C2 beaconing and behaviour.

The observed POST requests have a packet size that varies between 350 and 450 bytes; which confirms the potential malware is not just sending static beacons but also transmitting small chunks of data in each circle.

5. Evidence Collected

- **Malicious Domains:**
 1. dng-microsofts.com,

2. windows-msgas.com,
3. event-time-microsoft.org,
4. varying-rentals-calgary-predict.trycloudflare.com

- **Malicious IP address:**

1. 172.67.146.241,
2. 104.21.96.1,
3. 104.21.80.1,
4. 104.21.64.1,
5. 104.21.24.186
6. 104.21.16.1
7. 104.21.112.1
8. 104.16.231.132
9. 104.16.230.132

- **Wireshark Filter Used**

1. Dns
2. Ip.addr == {suspicious IP}
3. Http.request || tls.handshake.type eq 1

Follow TCP stream was also used to inspect the traffic content.

6. Indicator of Compromise

The following indicators suggested system compromise:

- DNS requests to malicious domains
- Suspicious outbound connections to malicious IPs
- Anomalous communication between internal host and external malicious IP
- GET request which contains obfuscated powershell codes to malicious domain which suggests potential malware payload delivery.
- Multiple structured POST requests with consistent time frame to malicious domain after the GET request which suggested potential C2 beaconing behaviour or data exfiltration.

7. Key Findings

- **Traffic type observed:** DNS,TCP,HTTP,HTTPS.
- **Malicious domains identified.**
- **Malicious IP communication detected**
- **Possible malware payload download observed**
- **Evidence of potential C2 activities.**

8. Conclusion and Recommended Actions

The PCAP analysis reveals multiple indicators of compromise which includes, malicious DNS resolution and multiple suspicious outbound communication to malicious external IP addresses which are consistent with command and control behaviours. The findings strongly suggest malware infection and post compromise activities on the affected hosts.

Recommended Actions

1. Block identified malicious IP addresses and domains at the Firewall and network security control.
2. Isolate affected endpoint from the network
3. Deep forensic on the affected hosts
4. Perform endpoint malware remediation and full system scanning
5. Review user activities and provide security awareness training if applicable.
6. Escalate Incident to Tier 2 SOC for deeper forensic analysis.